

* Review

1.3:

$$\bullet \lim_{x \rightarrow a} c = c \quad \text{and} \quad \lim_{x \rightarrow a} x = a$$

THEOREM 3.1

Suppose that $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ both exist and let c be any constant. The following then apply:

- (i) $\lim_{x \rightarrow a} [c \cdot f(x)] = c \cdot \lim_{x \rightarrow a} f(x)$,
- (ii) $\lim_{x \rightarrow a} [f(x) \pm g(x)] = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x)$,
- (iii) $\lim_{x \rightarrow a} [f(x) \cdot g(x)] = \left[\lim_{x \rightarrow a} f(x) \right] \left[\lim_{x \rightarrow a} g(x) \right]$ and
- (iv) $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} \quad \left(\text{if } \lim_{x \rightarrow a} g(x) \neq 0 \right)$.

THEOREM 3.2

For any polynomial $p(x)$ and any real number a ,

$$\lim_{x \rightarrow a} p(x) = p(a).$$

THEOREM 3.3

Suppose that $\lim_{x \rightarrow a} f(x) = L$ and n is any positive integer. Then,

$$\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)} = \sqrt[n]{L},$$

where for n even, we must assume that $L > 0$.

THEOREM 3.4

For any real number a , we have

- (i) $\lim_{x \rightarrow a} \sin x = \sin a$, (iii) if p is a polynomial and $\lim_{x \rightarrow p(a)} f(x) = L$,
- (ii) $\lim_{x \rightarrow a} \cos x = \cos a$ and then $\lim_{x \rightarrow a} f(p(x)) = L$.

$$f(p(a)) = L$$

$$\textcircled{1} \lim_{x \rightarrow a} (f(x))^n = \left[\lim_{x \rightarrow a} f(x) \right]^n \quad f(\lim_{x \rightarrow a} p(x)) = f(p(a))$$
$$\textcircled{2} \lim_{x \rightarrow a} x^n = a^n \quad L$$

THEOREM 3.5 (Squeeze Theorem)

Suppose that

$$f(x) \leq g(x) \leq h(x)$$

for all x in some interval (c, d) , except possibly at the point $a \in (c, d)$ and that

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L,$$

for some number L . Then, it follows that

$$\lim_{x \rightarrow a} g(x) = L, \text{ also.}$$

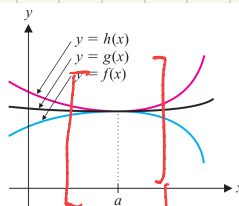


FIGURE 1.18
The Squeeze Theorem

Ex: Prove that $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$.

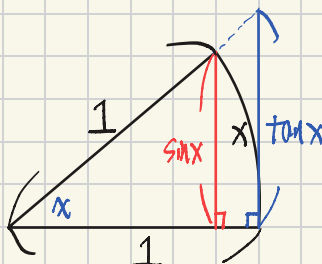
pf ① If $0 < x < \frac{\pi}{2}$, we obtain

$$\sin x < x < \tan x.$$

$$\text{Thus } 1 < \frac{x}{\sin x} < \frac{\tan x}{\sin x}$$

$$\Rightarrow 1 < \frac{x}{\sin x} < \frac{1}{\cos x}$$

$$\Rightarrow \cos x < \frac{\sin x}{x} < 1$$



② If $\frac{\pi}{2} < x < 0$, this implies $\frac{\pi}{2} > -x > 0$.

$$\text{By ①, we have } \cos(-x) < \frac{\sin(-x)}{(-x)} < 1$$

$$\Rightarrow \cos x < \frac{\sin x}{x} < 1$$

Since $\lim_{x \rightarrow 0} \cos x = 1$ and $\lim_{x \rightarrow 0} 1 = 1$,

by the Squeeze Theorem, $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$. //

Note:

$$\textcircled{1} \lim_{x \rightarrow 0} \frac{x}{\sin x} = \lim_{x \rightarrow 0} \frac{1}{\frac{\sin x}{x}} = \frac{1}{\lim_{x \rightarrow 0} \frac{\sin x}{x}} = \frac{1}{1} = 1$$

$$\textcircled{2} \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \lim_{x \rightarrow 0} \frac{2 \sin^2(\frac{x}{2})}{x^2} = \lim_{x \rightarrow 0} \left(\frac{\sin(\frac{x}{2})}{\frac{x}{2}} \right)^2 \cdot \frac{2}{4} = \frac{2}{4} = \frac{1}{2}$$

$$\begin{aligned} \sin^2 \theta &= \frac{1 - \cos 2\theta}{2} \\ \Rightarrow 2 \sin^2 \theta &= 1 - \cos 2\theta \\ \theta &= \frac{x}{2} \\ \Rightarrow 2 \sin^2 \frac{x}{2} &= 1 - \cos x \end{aligned}$$

$$10. \lim_{x \rightarrow 0} \frac{\tan x}{x} = \lim_{x \rightarrow 0} \frac{\sin x}{\cos x} \cdot \frac{1}{x}$$

$$= \lim_{x \rightarrow 0} \boxed{\frac{\sin x}{x}} \cdot \boxed{\frac{1}{\cos x}} = 1$$

$$11. \lim_{x \rightarrow 0} \frac{x \cos(x^2+1)}{x^2+x} = \lim_{x \rightarrow 0} \frac{\cancel{x} \cos(x^2+1)}{\cancel{x}(x+1)}$$

$$= \frac{\cos(0+1)}{0+1} = \cos 1$$

$$12. \lim_{x \rightarrow 0} x^2 \csc^2 x = \lim_{x \rightarrow 0} \frac{x^2}{\sin^2 x}$$

$$= \lim_{x \rightarrow 0} \left(\frac{x}{\sin x} \right)^2 = 1^2 = 1$$

$$14. \lim_{x \rightarrow 0} \frac{2x}{3 - \sqrt{x+9}} = \lim_{x \rightarrow 0} \frac{2x}{3 - \sqrt{x+9}} \cdot \frac{3 + \sqrt{x+9}}{3 + \sqrt{x+9}}$$

$$= \lim_{x \rightarrow 0} \frac{\cancel{2}x(3 + \sqrt{x+9})}{3^2 - (x+9)} \quad 9 - x - 9$$

$$= \lim_{x \rightarrow 0} \frac{\cancel{2}\cancel{x}(3 + \sqrt{x+9})}{-x}$$

$$= -2 \cdot (3 + \sqrt{9}) = -2 \cdot 6 = -12$$

$$17. \lim_{x \rightarrow 1} \left(\frac{1}{x-1} - \frac{2}{x^2-1} \right) = \lim_{x \rightarrow 1} \frac{(x+1)-2}{x^2-1}$$

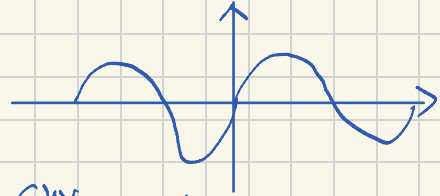
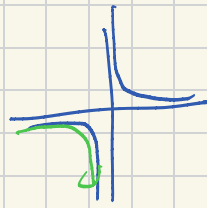
$$= \lim_{x \rightarrow 1} \frac{\cancel{x} + 1}{(x+1)\cancel{(x-1)}}$$

$$= \frac{1}{2}$$

$$16. \lim_{x \rightarrow 0} \left(\frac{2}{x} - \frac{2}{|x|} \right) = \text{DNE}$$

$$\lim_{x \rightarrow 0^-} \left(\frac{2}{x} + \frac{2}{x} \right) = \lim_{x \rightarrow 0^-} \frac{4}{x} = -\infty \quad (\text{DNE})$$

$$\lim_{x \rightarrow 0^+} \left(\frac{2}{x} - \frac{2}{x} \right) = 0$$



$$20. \lim_{x \rightarrow 0} \frac{|\sin x|}{x}$$

$$\lim_{x \rightarrow 0^-} \frac{|\sin x|}{x} = \lim_{x \rightarrow 0^-} \frac{-\sin x}{x} = -\lim_{x \rightarrow 0^-} \frac{\sin x}{x} = -1$$

$$\lim_{x \rightarrow 0^+} \frac{|\sin x|}{x} = \lim_{x \rightarrow 0^+} \frac{\sin x}{x} = 1$$

Since $\lim_{x \rightarrow 0^-} \frac{|\sin x|}{x} \neq \lim_{x \rightarrow 0^+} \frac{|\sin x|}{x}$, $\lim_{x \rightarrow 0} \frac{|\sin x|}{x}$ does not exist.

$$27. \lim_{x \rightarrow 2} \frac{\sin(x^2-4)}{x^2-4}$$

$$\lim_{t \rightarrow 0} t = 0$$

Let $t = x^2 - 4$, $x \rightarrow 2 \Rightarrow t \rightarrow 0$

$$\lim_{x \rightarrow 2} \frac{\sin(x^2-4)}{x^2-4} = \lim_{t \rightarrow 0} \frac{\sin t}{t} = 1$$

$$\lim_{x \rightarrow 0} \frac{\sin ax}{ax} = 1$$

$$\Rightarrow \frac{\sin 3x}{3x}$$

$$\begin{aligned} 28. \lim_{x \rightarrow 0} \frac{\tan 3x}{5x} &= \lim_{x \rightarrow 0} \frac{\sin 3x}{\cos 3x} \cdot \frac{1}{5x} \\ &= \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \cdot \frac{3}{5 \cos 3x} \\ &= 1 \cdot \frac{3}{5} = \frac{3}{5} \end{aligned}$$

$$\text{If } a \neq b, \lim_{x \rightarrow 0} \frac{\sin ax}{bx} = ? \quad \frac{a}{b}$$

32. Suppose that $f(x)$ is bounded: that is, there exists a constant M such that $|f(x)| \leq M$ for all x . Use the Squeeze Theorem to prove that $\lim_{x \rightarrow 0} x^2 f(x) = 0$.

$$|f(x)| \leq M \Leftrightarrow -M \leq f(x) \leq M$$

pf Since $|f(x)| \leq M$ for all x ,

$$-M \leq f(x) \leq M$$

$$\Rightarrow -Mx^2 \leq x^2 f(x) \leq Mx^2$$

$\lim_{x \rightarrow 0} -Mx^2 = 0$, $\lim_{x \rightarrow 0} Mx^2 = 0$, by the Squeeze Thm, $\lim_{x \rightarrow 0} x^2 f(x) = 0$. //

Thm

Suppose that $f(x)$ is a bounded function, that is $\exists M \in \mathbb{R}$ st $|f(x)| \leq M$. If $g(x)$ is a function and $\lim_{x \rightarrow 0} g(x) = 0$, then

$$\lim_{x \rightarrow 0} f(x)g(x) = 0.$$

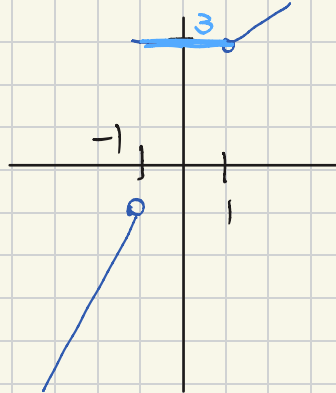
23. $\lim_{x \rightarrow -1} f(x)$, where $f(x) = \begin{cases} 2x + 1 & \text{if } x < -1 \\ 3 & \text{if } -1 < x < 1 \\ 2x + 1 & \text{if } x > 1 \end{cases}$

24. $\lim_{x \rightarrow 1} f(x)$, where $f(x) = \begin{cases} 2x + 1 & \text{if } x < -1 \\ 3 & \text{if } -1 < x < 1 \\ 2x + 1 & \text{if } x > 1 \end{cases}$

$$\lim_{x \rightarrow -1^-} f(x) = \lim_{x \rightarrow -1^-} 2x + 1 = -2 + 1 = -1$$

$$\lim_{x \rightarrow -1^+} f(x) = \lim_{x \rightarrow -1^+} 3 = 3$$

Since $\lim_{x \rightarrow -1^-} f(x) \neq \lim_{x \rightarrow -1^+} f(x)$, $\lim_{x \rightarrow -1} f(x)$ does not exist.



In exercises 33–36, use the given position function $f(t)$ to find the velocity at time $t = a$.

33. $f(t) = t^2 + 2$, $a = 2$

34. $f(t) = t^2 + 2$, $a = 0$

35. $f(t) = t^3$, $a = 0$

36. $f(t) = t^3$, $a = 1$

33. $V = \lim_{t \rightarrow 2} \frac{f(t) - f(2)}{t - 2}$

$$= \lim_{t \rightarrow 2} \frac{t^2 + 2 - 6}{t - 2} = \lim_{t \rightarrow 2} \frac{t^2 - 4}{t - 2}$$

$$= \lim_{t \rightarrow 2} \frac{(t+2)(t-2)}{t-2} = 2+2 = 4$$

$f'(t)$: Chapter 2

Just focus on

$$u = \frac{\Delta f}{\Delta t} = \frac{f(t_f) - f(t_i)}{t_f - t_i}$$